Paper id: 250621

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VI) THEORY EXAMINATION 2024-25
MACHINE LEARNING TECHNIQUES

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define Machine Learning and its importance.	1	K1
b.	What is the difference between linear and logistic regression?	2	K2
c.	Define EM Algorithm in the context of Bayesian Learning.	2	K2
d.	What is inductive bias in decision tree learning, and why is it necessary for generalization?	3	K1
e.	What is the role of an activation function in neural networks?	4	K2
f.	What is the purpose of pooling layers in CNN?	4	K2
g.	Differentiate between crossover and mutation.	5	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain different types of Machine Learning with examples.	1	K2
b.	Describe the working of the Naive Bayes classifier with a numerical example.	2	K3
c.	Discuss the role of entropy and information gain in building decision trees.	3	K4
d.	Explain the working principle of the Self-Organizing Map (SOM) algorithm. Describe how the neurons are organized, how the best matching unit (BMU) is identified, and how weights are updated.	4	K2
e.	Define a Markov Decision Process (MDP) and explain its key components. How does the MDP framework support decision-making in reinforcement learning?	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Discuss the components involved in designing a learning system.	1	K2
b.	Compare various ML approaches such as ANN, SVM, and Decision Trees.	1	K4

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Given the dataset: X: [1, 2, 3, 4, 5] Y: [2, 4, 5, 4, 5] Perform linear regression using the least squares method and find the regression coefficients.	2	K3
b.	Given two classes: Class A: (1,2), (2,3), (3,3) Class B: (6,5), (7,8), (8,8) Using a linear kernel SVM, determine the separating hyperplane and calculate the margin.	2	K3

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5. Attempt any *one* part of the following: 07 x 1 = 07

a.	Classify the test point (3, 3) using k-NN (k=3) with training data: (1,1)-A, (2,2)-A, (3,4)-B, (5,5)-B	3	K3
b.	Explain the ID3algorithm in detail. Discuss how it uses entropy and information gain to construct a decision tree. Analyze the strengths and limitations of the ID3 algorithm in the context of decision tree learning.	3	K2

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Discuss the derivation of the backpropagation algorithm.	4	K2
b.	Explain the architecture and working principles of a Convolutional Neural Network (CNN).	4	K3

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Describe the Q-learning algorithm with a step-by-step example.	5	K3
b.	Explain the cycle of reproduction in Genetic Algorithms.	5	K3

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